CONCEPTUAL SERIES
Amazon DynamoDB

Name: ________________ Cohort/Batch: ________________ Date: ________________ Score: ________

AMAZON DYNAMODB PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Databases

TOTAL: 10 QUESTIONS

Q1 What type of database is Amazon DynamoDB and what are its core strengths? *Threat Model*

Q2 How does Amazon DynamoDB guarantee consistency and handle transactions? *Architecture*

Q3 What index types does Amazon DynamoDB support and when do you use each? *Configuration*

Q4 How do you design a schema or data model in Amazon DynamoDB? *Hardening*

Q5 How does Amazon DynamoDB scale horizontally? *SSO / IdP*

Q6 How do you monitor and tune slow queries in Amazon DynamoDB? *Observability*

Q7 What backup and restore strategies does Amazon DynamoDB support? *Lifecycle*

Q8 How do you handle schema migrations in Amazon DynamoDB? *Compliance*

Q9 What security features does Amazon DynamoDB provide? *Testing*

Q10 What are common anti-patterns when using Amazon DynamoDB in production? *Tuning*

ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Amazon DynamoDB is a relational, document,

Mitigation

Amazon DynamoDB is a relational, document, key-value, graph, or time-series store. Its strengths such as ACID guarantees, horizontal scale, or flexible schema guide the workloads it is best suited for.

A2 Amazon DynamoDB provides either full ACID

Primitives

Amazon DynamoDB provides either full ACID transactions or eventual consistency depending on its CAP theorem positioning. Transaction support varies from none (simple K/V stores) to serializable isolation (relational DBs).

A3 Amazon DynamoDB offers B-tree, hash, full-text,

Hardening

Amazon DynamoDB offers B-tree, hash, full-text, spatial, or composite indexes depending on the engine. Choose based on query patterns: B-tree for range queries, hash for equality lookups, full-text for search.

A4 Schema design in Amazon DynamoDB involves

Gotchas

Schema design in Amazon DynamoDB involves normalizing relations (relational DB) or denormalizing for access patterns (document/key-value). Understanding query needs upfront prevents costly migrations later.

A5 Amazon DynamoDB scales via replication (read

Federation

Amazon DynamoDB scales via replication (read replicas) for read-heavy workloads and sharding (partitioning data by key) for write-heavy ones. Some Amazon DynamoDB implementations handle this automatically; others require manual configuration.

A6 Amazon DynamoDB provides a slow query

Observability

Amazon DynamoDB provides a slow query log, explain/explain plan, or profiling tools to surface inefficient queries. Common fixes include adding indexes, rewriting queries, or increasing cache size.

A7 Amazon DynamoDB supports logical backups

Automation

Amazon DynamoDB supports logical backups (export SQL or BSON), physical backups (filesystem snapshot), and continuous replication to a standby. Point-in-time recovery requires WAL/oplog archiving on top of backups.

A8 Schema migrations in Amazon DynamoDB are

Governance

Schema migrations in Amazon DynamoDB are managed with versioned migration files applied in order by a migration tool. Migrations should be idempotent and tested in staging before production to avoid downtime.

A9 Amazon DynamoDB supports role-based

Assessment

Amazon DynamoDB supports role-based access control, encrypted connections (TLS), encryption at rest, and audit logging. Always restrict user privileges to the minimum needed and disable anonymous access in production.

A10 Common mistakes include selecting all

Optimization

Common mistakes include selecting all columns instead of needed fields, missing indexes on frequently queried columns, running migrations without backups, and storing large blobs directly in the database instead of object storage.